Impact of Payroll on Major League Baseball Team Success

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Abstract

This paper examines the relationship between Major League Baseball (MLB) team payroll distribution and team success over a ten-season period (2015–2024), considering both overall performance and year-to-year consistency. In recent seasons, record-breaking free agent contracts have raised questions about whether a team can simply buy their way to more wins. During this period, there has been a gradual increase in overall team payrolls, accompanied by a widening gap between the highest and lowest-spending teams. This growing disparity raises concerns about competitive balance and whether financial constraints hinder smaller-market teams' ability to succeed. To explore the validity of the concern, this paper considers all players who appeared in at least one game for a team in a given season and categorizes them into three payroll groups: pre-arbitration, arbitration-eligible, and veteran players. The analysis examines how the allocation of payroll across these groups influences team performance and consistency over time. Specifically, this paper seeks to answer: Can small-market teams achieve the same level of success as large-market teams despite payroll disparities? How does the distribution of payroll across different player categories impact a team's overall performance and year-to-year variation in wins? The findings aim to contribute to the broader discussion on competitive balance in Major League Baseball and the effectiveness of different roster-building strategies in achieving success as well as sustaining success.

Introduction

Baseball is a game of numbers and strategy where teams continuously seek to gain any competitive advantage they can find. One such area of focus for a team is roster construction. Major League Baseball (MLB) is unique compared to most other professional sports leagues as it has its own development system with minor league affiliates. Teams must balance their focus between putting the best possible product on the field at the major league level while also working on development in the minor leagues to hopefully help with the team's future major league success. A team's major league roster is constantly changing whether it be due to injuries, performance fluctuations, or financial constraints, making the task of managing a roster quite difficult.

The financial aspect of roster management has become more significant in recent seasons with numerous record-breaking contracts. In the 2023 offseason, Shohei Ohtani signed a historic deal for $700 million over 10 seasons with the Los Angelos Dodgers (Adler, 2025). Prior to Ohtani's deal the highest paid player was Mike Trout who signed a 12-year $465.5 million contract extension with the Los Angelos Angels of Anaheim before to the 2019 season (Haley, 2024). The deal with Ohtani and the Dodgers proved to be immediately successfully as Ohtani won the National League MVP award and helped lead the Dodgers to a World Series title in the first season of the contract. However, his record contract was short-lived as Juan Soto signed a 15-year $765 million contract with the New York Mets in the following offseason (Adler, 2025).

These high-profile contracts raise important questions about the relationship between payroll and team success. Can a team simply buy its way to victory? Moreover, the financial disparities between large-market franchises, such as those in Los Angelos and New York, and smaller-market teams highlight concerns about competitive balance. With differing payroll budgets, smaller-market teams must find alternative strategies to remain competitive.

This study investigates not just the amount a team spends, but how that spending is allocated across different player categories. MLB players are generally classified as pre-arbitration, arbitration-eligible, or veteran based on their years of service, with each group commanding different salary levels and contributing in different ways. By examining the relationship between payroll composition, team market size, and on-field performance, this research aims to provide a deeper understanding of the strategic choices teams make and the impact of those choices on competitive outcomes.

Background

MLB Rosters

A Major League Baseball team's roster is made up of many different types of players all with different contracts. The 40-man roster consists of players on the team signed to a major league contract ("MLB Roster History," n.d.). It includes those players on the active roster, the injured list (except the 60-day IL), and those in minor leagues that have a major league contract. A team is allowed to have 26 players on their active roster (before 2020 it was only 25) which are the players currently eligible to play for the team. Throughout the season, a team's active roster and 40-man go through many changes. While on a team's active roster (or injured list), a player earns major league service time.

A player can come to be on a team's roster in a few different ways. First is the First Year Player Draft where teams select U.S. high school and college players ("Transaction Glossary," n.d.). Once a player reaches six years of service time with a club, they are eligible for free agency. Players with fewer than six years of service but were released by the club are also eligible. In free agency, a player can sign with any team with which the two sides reach an agreement on a contract. A player can sign either a Major League or a Minor League deal with a team. The player is assigned to the 40-man roster when they sign a Major League deal. There is also international free agency where players follow into a few different categories. There are foreign professional players that have played in leagues in other countries such as Japan, Korea, Tawain, Mexico, or Cuba. There are also amateur players that must at least 16 years of age at the time of signing and turn 17 either by

September 1st or by the end of their first professional season. Another common method of acquiring a player is through a trade. Between the end of the regular season and the trade deadline (between July 28th and August 3rd), players can be traded between teams that can reach an agreement. Additionally, players can be claimed off waivers or selected in the Rule 5 Draft.

Player Types

MLB players can be classified into three service time–based categories that correspond to different stages of career development and salary negotiation rights: pre-arbitration, arbitration-eligible, and veteran. Pre-arbitration players have fewer than three years of MLB service time and are not yet eligible for salary arbitration. These players generally earn salaries near the league minimum and represent the least costly segment of a team's roster. Arbitration-eligible players have between three and six years of service (or qualify under "Super Two" status) and can negotiate salaries through the arbitration process if they do not reach an agreement with their team. Finally, veteran players possess six or more years of service time, making them eligible for free agency and typically commanding the highest salaries due to their experience and established performance.

Market Size

A team's market size refers to the scale of its media market and overall revenue-generating potential, which often influences financial flexibility and roster-building strategies. Teams are commonly grouped into small, mid, or large-market categories based on factors such as regional population, local broadcasting reach, sponsorship opportunities, and ticket sales. Large-market teams are often located in densely populated metropolitan areas and tend to benefit from higher revenue streams making them more capable of sustaining large payrolls. In contrast, small-market teams typically face budgetary constraints and may prioritize cost-effective player development and payroll efficiency. Understanding market size is essential when evaluating team investment strategies, as it helps contextualize differences in spending behavior and competitive advantages across the league.

Literature Review

The question of how teams can build competitive rosters within budget constraints has been a persistent focus in baseball research, particularly since the publication of Michael Lewis's *Moneyball* in 2003. Hakes and Sauer (2006) provide empirical support for the Moneyball thesis, showing that the early-2000s labor market undervalued on-base percentage (OBP) relative to slugging percentage. Their analysis revealed that OBP contributed more significantly to team wins, yet players with higher OBP earned far less than power hitters. This inefficiency enabled budget-conscious teams like the Oakland Athletics to remain competitive by targeting undervalued skills. However, the market corrected itself rapidly after Lewis's book popularized these findings, illustrating how quickly competitive advantages in roster construction can disappear once widely adopted.

Beyond individual player valuation, researchers have explored how the distribution of payroll within a team influences performance. Annala and Winfree (2011) find that internal payroll inequality, measured using Gini coefficients, has a negative impact on team success, independent of total spending. This suggests that even high-payroll teams may suffer if compensation is overly concentrated among a few stars, potentially disrupting team cohesion. Their findings point to the strategic importance of how teams allocate payroll across players, not just how much they spend overall.

Building on this theme, recent work has begun to integrate predictive and prescriptive analytics to optimize roster construction under financial constraints. Barnes et al. (2024) develop a regression-based optimization model that incorporates performance projections, salary data, and positional flexibility to identify cost-effective player combinations. Their findings suggest that investing in several moderately priced, high-value players often yields greater team success than acquiring a few marquee stars, especially in years with limited elite talent. This supports a more nuanced view of spending efficiency that aligns closely with this paper's focus on payroll composition rather than just payroll size.

Complementary studies using Data Envelopment Analysis (DEA) offer broader perspectives on efficiency in professional sports. Einolf (2004) compare the NFL and MLB, showing that NFL teams tend to be more efficient due to mechanisms like revenue sharing and salary caps. MLB teams, especially those in large markets, often spend more without proportionate returns. Similarly, Lewis et al. (2007) find that small-market MLB teams face structural disadvantages in achieving competitive parity, even when spending efficiently. They introduce a two-stage DEA model to assess how payroll translates into talent and wins, identifying overspending or underperformance relative to an estimated minimum salary threshold. These findings reinforce concerns raised in the introduction about competitive imbalance in MLB and point to the importance of optimizing not just how much is spent, but how those resources are allocated across the roster.

While prior research has made significant contributions to our understanding of player valuation, payroll distribution, and team efficiency, there remains a gap in exploring how the composition of payroll, specifically the distribution across player service time categories, relates to on-field success. This study builds on the existing literature by splitting team payrolls into pre-arbitration, arbitration-eligible, and veteran player portions, examining how different spending strategies manifest across teams of varying market sizes. By focusing on how payroll is structured rather than simply how much is spent, this research seeks to provide more actionable insights for roster construction in a financially imbalanced league.

Data

Data Collection

Data collection focused on two main areas: team performance and player-level payroll. Team performance data and forty-man roster information were sourced from Baseball Reference ("MLB Stats, Scores, History, Records," n.d.). The forty-man roster included all players who appeared in at least one game for their respective teams in a given season. Payroll data was obtained from Spotrac ("MLB," n.d.), which provides detailed information on each team's total payroll, including breakdowns by player roster designation for each season. All data was collected for the 2015 through the 2024 seasons. However, the 2020 season was excluded from the analysis as the season was shorten to 60 games instead of the typical 162 games due to the COVID-19 pandemic.

Data Cleaning

In addition to the team data from Baseball Reference, several supplementary variables were incorporated. A dataset was constructed to include each team's league, division, and market size, with market size classifications taken from *Baseball Almanac* (see Table A1). This information was merged with the team performance dataset. A variable for win variability was created by calculating the standard deviation of a team's win totals over a three-year window (the two preceding seasons plus the current season).

To integrate the payroll and roster data, several data cleaning steps were required. The Spotrac payroll data classified players by roster designation (e.g., active roster, injured list, retained, deferred, minor league) as of the season's end. Some players appeared multiple times on the same team's roster with different designations. To avoid duplication, those entries were consolidated by summing their payroll amounts to reflect the total salary paid by that team. A single player could still appear multiple times across the dataset if they played for more than one team during the season.

Matching players between the Baseball Reference and Spotrac datasets involved a multi-step process to ensure complete payroll coverage for all players who appeared in a Major League game. The initial match used player name, position (pitcher or hitter), team, and year, successfully matching 97.2% of records. Unmatched cases typically resulted from discrepancies in naming conventions (e.g., use of nicknames). A second matching step using last name, position, team, and year resolved all but 0.01% of cases. A final pass matched players by name, team, and year (ignoring position discrepancies) leaving only 0.009% of players unmatched. These were players that appeared in at least one game, but Spotrac did not have record of their contract for the season.

A key variable in the analysis was the player's status, defined as pre-arbitration, arbitration-eligible, or veteran. This information was primarily sourced from Spotrac, but approximately 21% of records were missing status values. When available, status was imputed using years of MLB service time. For players lacking both status and service time, additional imputation strategies were used. If the player appeared in the dataset with a different team in the same season and had a valid status in that row, the value was transferred. These cases typically involved players who were traded or claimed off waivers. If a status was still missing, the value was backfilled using the previous season's status. The remaining unmatched cases were generally confirmed to be pre-arbitration players. The only players left without a status value were those for whom payroll data was entirely missing.

After cleaning and merging the player-level and team-level datasets, a summary dataset was created. This final dataset included each team's total payroll for the season, as well as payroll distributions by player status.

Exploratory Data Analysis

Figure 1 illustrates the relationship between team total payroll and the number of regular season wins for each year from 2015 to 2024. Across all seasons, there is a positive correlation between payroll and wins; however, the strength of this correlation has diminished in more recent years. Notably, in 2023, two outliers, the Baltimore Orioles and Tampa Bay Rays, achieved 101 and 99 wins, respectively, despite having total payrolls under $100 million which affected the magnitude of the correlation. Additionally, there has been a gradual increase in overall team payrolls during this period, accompanied by a widening gap between the highest- and lowest-spending teams.

Figure 2 presents the average payroll allocation and player distribution for teams from 2015 through 2024, organized by market size. Pre-arbitration players make up

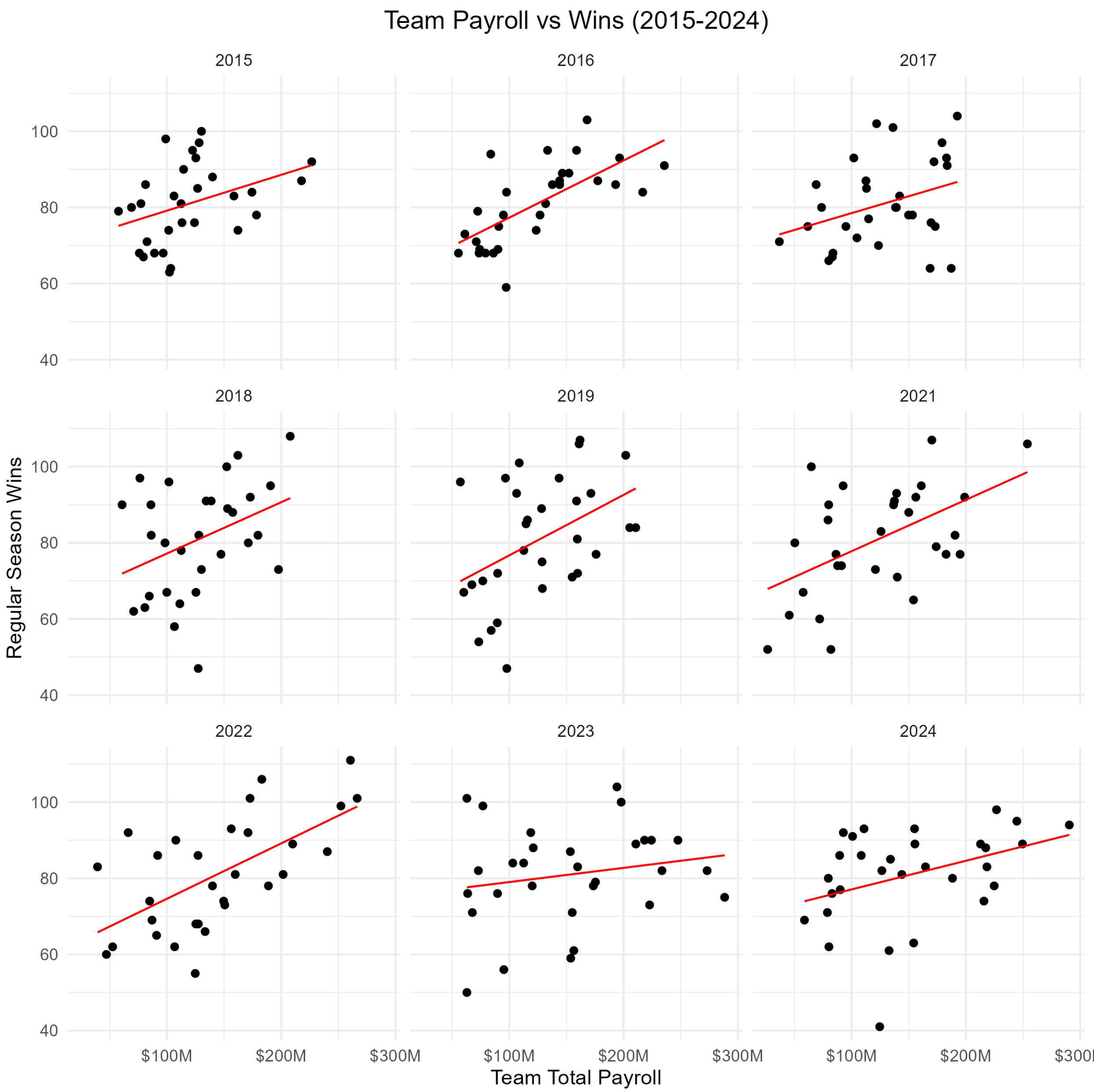

Figure 1

Relationship Between Team Total Payroll and Wins approximately half of the players who appear for a team in a given season but account for only a small proportion of total payroll. Also, the distribution of payroll allocations is not uniform for the different market sizes.

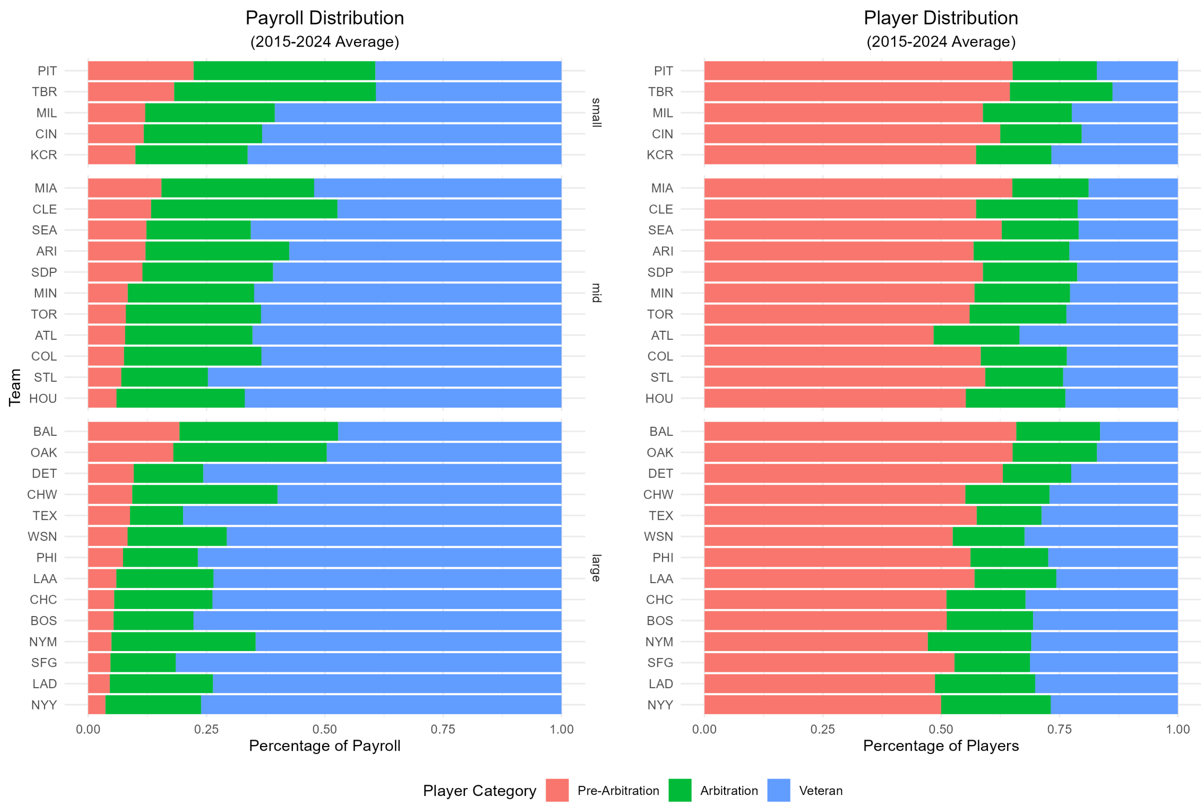

Figure 2

Average Team Payroll Allocation and Player Distribution

Figure 3 further investigates the relationship between market size, team performance, and payroll characteristics. It displays the distributions for several metrics: regular season wins, total payroll, payroll Gini index, and the percentage of total payroll allocated to pre-arbitration, arbitration-eligible, and veteran players. P-values from ANOVA tests are reported for differences in means across market sizes. Regular season wins is the only metric for which the difference by market size is not statistically significant ($p > 0.05$). However, significant differences are observed in payroll levels and payroll distributions. Large-market teams have the highest average payrolls and the highest payroll Gini indices, indicating greater disparities between their highest- and lowest-paid players. Large-market teams also allocate a greater share of payroll to veteran players. Small- and mid-market teams rely more heavily on pre-arbitration and arbitration-eligible players, with small-market teams particularly dependent on pre-arbitration talent.

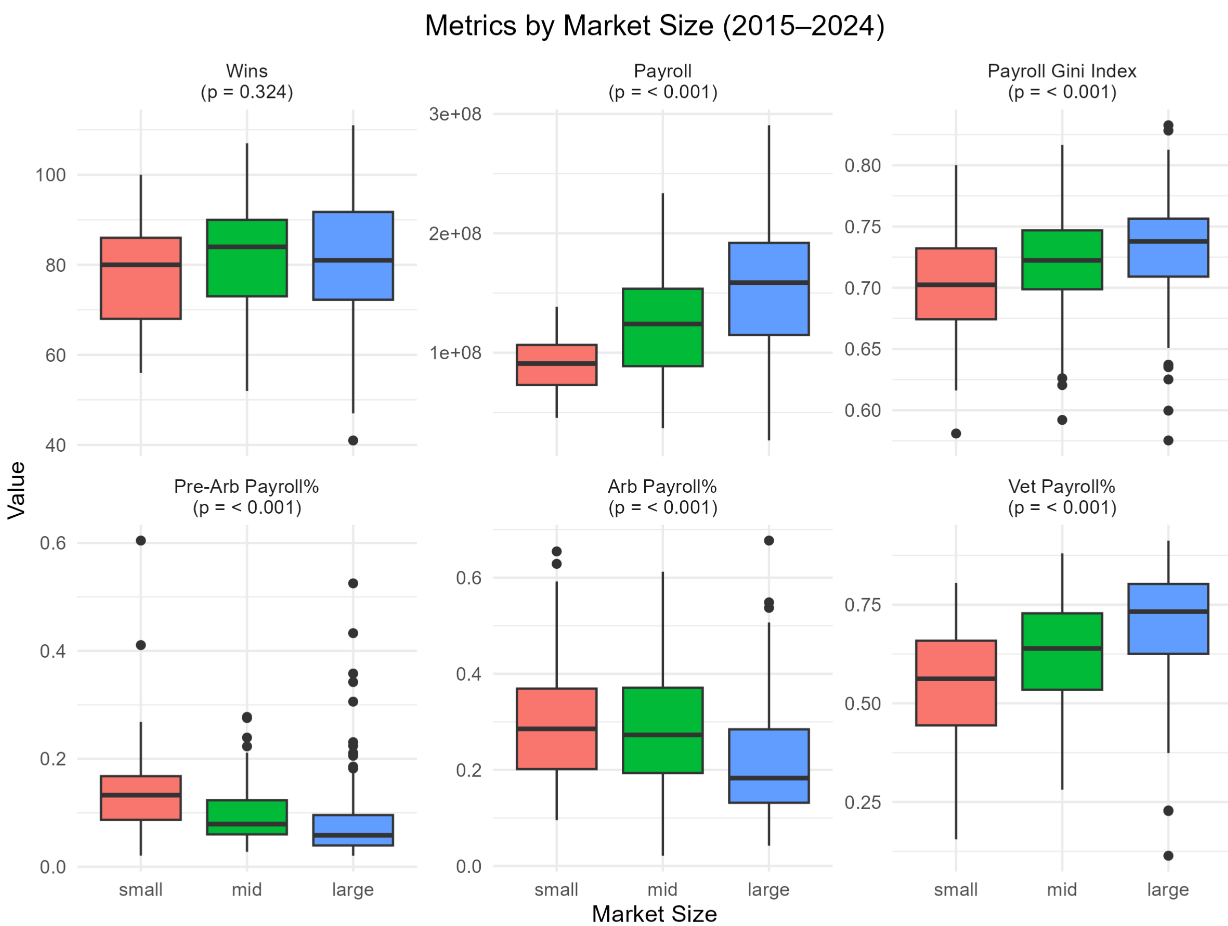

Figure 3

Distribution of Wins and Payroll Metrics by Market Size

Figure 4 examines team performance consistency over the ten-season span from 2015 to 2024. The Los Angeles Angels were the most consistent team in terms of wins, with a standard deviation of 6.28 wins per season, although they consistently performed below average, averaging approximately 75 wins per season. Conversely, the Baltimore Orioles were the most inconsistent team, with a standard deviation of 19.31 wins. The

Orioles number of wins in a season ranged from 47 wins in 2018 to 101 wins in 2023. A negative correlation is observed between a team's average wins per season and the standard deviation of its wins, suggesting that teams with higher average win totals tend to be more consistent year to year. A similar negative relationship exists between average payroll and win variability, implying that teams with higher payrolls typically demonstrate more consistent performance. However, several large-market teams deviate from this trend, exhibiting higher variability in performance relative to their payroll levels. Lastly, a slight positive correlation is observed between the standard deviation of payroll and the standard deviation of wins, suggesting that greater fluctuations in payroll are associated with greater variability in on-field performance. Smaller-market teams tend to have lower payrolls and less payroll variability, yet their win variability remains similar to that of larger-market teams.

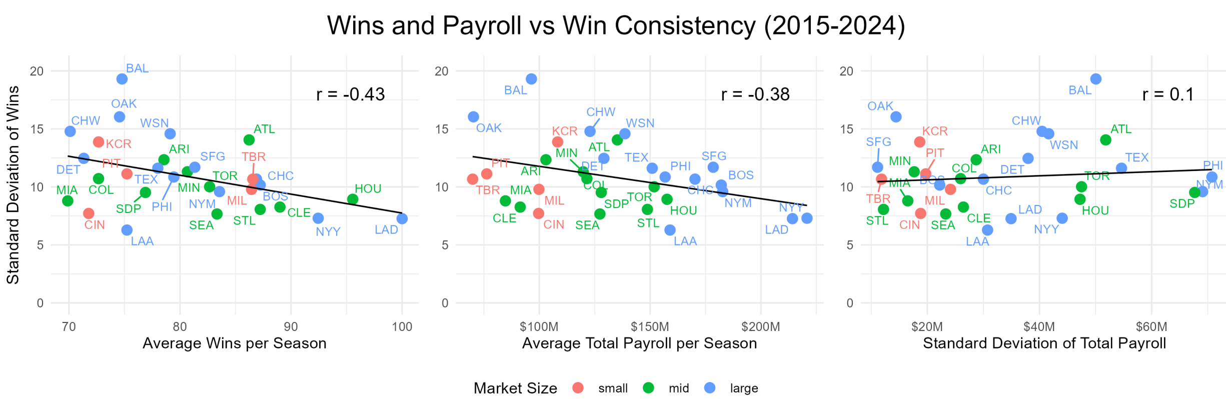

Figure 4

Distribution of Wins and Payroll Metrics by Market Size

Model

To examine the relationship between team payroll and regular-season performance, a series of linear regression models were estimated in which the dependent variable is the number of team wins in a given season. The primary independent variable is total team payroll, expressed in millions of dollars. Additionally, log-linear models were estimated in which the natural logarithm of payroll is used to capture percentage changes in spending. Across specifications, fixed effects were progressively added to control for unobserved heterogeneity at the team and year levels.

Linear Models

Baseline

The baseline model estimates a pooled ordinary least squares (OLS) regression:

$$\text{wins}_{it} = \beta_0 + \beta_1 \cdot \text{total\_payroll\_m}_{it} + \epsilon_{it} \tag{1}$$

This model tests whether higher payroll is associated with more wins, without accounting for differences across teams or over time.

Team Fixed Effects

To account for unobserved, time-invariant differences between teams (e.g., organizational strategy, market size, or managerial quality) a team fixed effects model is estimated:

$$\text{wins}_{it} = \beta_1 \cdot \text{total\_payroll\_m}_{it} + \alpha_i + \epsilon_{it} \tag{2}$$

Here, α_i captures team-specific factors that do not vary over time.

Year Fixed Effects

Next, to account for shocks or structural changes that affect all teams in a given year (e.g., inflation, rule changes, overall competitiveness), a year fixed effects model is estimated:

$$\text{wins}_{it} = \beta_1 \cdot \text{total\_payroll\_m}_{it} + \gamma_t + \epsilon_{it} \tag{3}$$

Two-Way Fixed Effects

Finally, a two-way fixed effects model includes both team and year fixed effects, providing the most rigorous control for unobserved heterogeneity:

$$\text{wins}_{it} = \beta_1 \cdot \text{total\_payroll\_m}_{it} + \alpha_i + \gamma_t + \epsilon_{it} \tag{4}$$

This specification allows for differences in both team characteristics and year-specific contexts, yielding a more robust estimate of the relationship between payroll and wins.

Log-Linear Models

To explore whether percentage changes in payroll are associated with changes in team wins, log-linear models were estimated, where the independent variable is the log transformation of total payroll (in millions).

Baseline

The baseline log-linear model is:

$$\text{wins}_{it} = \beta_0 + \beta_1 \cdot \log(\text{total\_payroll\_m}_{it}) + \epsilon_{it} \tag{5}$$

Team Fixed Effects

$$\text{wins}_{it} = \beta_1 \cdot \log(\text{total\_payroll\_m}_{it}) + \alpha_i + \epsilon_{it} \tag{6}$$

Year Fixed Effects

$$\text{wins}_{it} = \beta_1 \cdot \log(\text{total\_payroll\_m}_{it}) + \gamma_t + \epsilon_{it} \tag{7}$$

Team and Year Fixed Effects

$$\text{wins}_{it} = \beta_1 \cdot \log(\text{total\_payroll\_m}_{it}) + \alpha_i + \gamma_t + \epsilon_{it} \tag{8}$$

These models assess whether a percentage increase in payroll is associated with a change in team performance, controlling for relevant sources of unobserved heterogeneity.

Payroll Allocation Models

To further explore the impact of payroll structure on team performance, additional models were estimated using the share of payroll allocated to different player types: pre-arbitration, arbitration-eligible, and veteran players. These variables represent the percentage of total payroll allocated to each player group in a given season. The sum of the three categories equals 100% for each team.

Baseline

$$\text{wins}_i = \beta_1 \cdot \text{pre\_arb\_payroll\_pct}_i + \beta_2 \cdot \text{arb\_payroll\_pct}_i + \beta_3 \cdot \text{vet\_payroll\_pct}_i + \epsilon_i \quad (9)$$

This model examines the relationship between the distribution of payroll across player types and team wins, excluding an intercept to avoid perfect multicollinearity given the compositional nature of the percentages.

Total Payroll Control

$$\begin{aligned} \text{wins}_i = {} & \beta_1 \cdot \text{pre\_arb\_payroll\_pct}_i + \beta_2 \cdot \text{arb\_payroll\_pct}_i + \beta_3 \cdot \text{vet\_payroll\_pct}_i \\ & + \beta_4 \cdot \text{total\_payroll\_m}_i + \epsilon_i \end{aligned} \quad (10)$$

This model adds total payroll (in millions) to assess the joint effects of payroll amount and its allocation.

Total Payroll Interaction

$$\begin{aligned} \text{wins}_i = {} & \beta_1 \cdot \text{pre\_arb\_payroll\_pct}_i + \beta_2 \cdot \text{arb\_payroll\_pct}_i + \beta_3 \cdot \text{vet\_payroll\_pct}_i \\ & + \beta_4 \cdot \text{total\_payroll\_m}_i \\ & + \beta_5 \cdot (\text{arb\_payroll\_pct}_i \times \text{total\_payroll\_m}_i) \\ & + \beta_6 \cdot (\text{vet\_payroll\_pct}_i \times \text{total\_payroll\_m}_i) + \epsilon_i \end{aligned} \quad (11)$$

Interaction terms allow the marginal effect of payroll allocation to vary depending on total payroll, capturing potential complementarities or diminishing returns.

Market Size Control

$$\text{wins}_i = \beta_1 \cdot \text{pre\_arb\_payroll\_pct}_i + \beta_2 \cdot \text{arb\_payroll\_pct}_i + \beta_3 \cdot \text{vet\_payroll\_pct}_i + \beta_4 \cdot \text{market\_size}_i + \epsilon_i \tag{12}$$

This model includes market size as a control to account for external constraints or advantages faced by teams in different markets.

Win Variability Models

To assess the stability of team performance, a secondary outcome variable was considered: win variability, defined as the standard deviation of a team's wins over a three-year window (the two preceding seasons plus the current season). This captures the consistency of a team's performance, which may be influenced by the way payroll is allocated.

Baseline

$$\text{win\_variability}_i = \beta_1 \cdot \text{pre\_arb\_payroll\_pct}_i + \beta_2 \cdot \text{arb\_payroll\_pct}_i + \beta_3 \cdot \text{vet\_payroll\_pct}_i + \epsilon_i \tag{13}$$

Total Payroll Control

$$\begin{aligned} \text{win\_variability}_i = {} & \beta_1 \cdot \text{pre\_arb\_payroll\_pct}_i + \beta_2 \cdot \text{arb\_payroll\_pct}_i + \beta_3 \cdot \text{vet\_payroll\_pct}_i \\ & + \beta_4 \cdot \text{total\_payroll\_m}_i + \epsilon_i \end{aligned} \tag{14}$$

Total Payroll Interaction

$$
\begin{aligned}
\text{win\_variability}_i = {} & \beta_1 \cdot \text{pre\_arb\_payroll\_pct}_i + \beta_2 \cdot \text{arb\_payroll\_pct}_i + \beta_3 \cdot \text{vet\_payroll\_pct}_i \\
& + \beta_4 \cdot \text{total\_payroll\_m}_i \\
& + \beta_5 \cdot (\text{arb\_payroll\_pct}_i \times \text{total\_payroll\_m}_i) \\
& + \beta_6 \cdot (\text{vet\_payroll\_pct}_i \times \text{total\_payroll\_m}_i) + \epsilon_i
\end{aligned}
\tag{15}
$$

Market Size Control

$$
\begin{aligned}
\text{win\_variability}_i = {} & \beta_1 \cdot \text{pre\_arb\_payroll\_pct}_i + \beta_2 \cdot \text{arb\_payroll\_pct}_i + \beta_3 \cdot \text{vet\_payroll\_pct}_i \\
& + \beta_4 \cdot \text{market\_size}_i + \epsilon_i
\end{aligned}
\tag{16}
$$

These models help determine whether certain payroll allocation strategies are associated with more consistent performance, even after accounting for total spending or market context.

Results

Effect of Total Payroll on Team Wins

Table 1 presents the results from linear and log-linear regressions estimating the effect of total payroll on regular season wins corresponding to Equations 1–8.

The baseline linear model shows a strong positive relationship between total payroll and team wins. For each additional million dollars spent on payroll, teams gained approximately 0.101 wins ($\beta = 0.101$, $p < 0.001$). The model's intercept indicates that a hypothetical team with zero payroll would be expected to win about 67.5 games ($\beta_0 = 67.532$, $p < 0.001$). This baseline model explains approximately 17% of the variance in team wins ($R^2 = 0.169$), suggesting that while payroll is a significant predictor of performance, other factors also substantially influence team success.

Table 1

Effect of Total Payroll on Regular Season Wins

	Linear				**Log**			
	Baseline	Team FE	Year FE	Team + Year FE	Baseline	Team FE	Year FE	Team + Year FE
(Intercept)	67.532***				19.607*			
	(1.960)				(8.302)			
total_payroll_m	0.101***	0.094***	0.107***	0.106***				
	(0.014)	(0.017)	(0.017)	(0.021)				
log(total_payroll_m)					12.764***	13.120***	13.240***	14.223***
					(1.720)	(2.111)	(1.609)	(2.675)
Num. Obs.	270	270	270	270	270	270	270	270
R^2	0.169	0.406	0.178	0.414	0.170	0.423	0.177	0.430
Adj. R^2	0.166	0.331	0.150	0.318	0.167	0.351	0.148	0.336
AIC	2106.1	2073.6	2119.0	2085.7	2105.6	2065.6	2119.6	2078.3
RMSE	11.87	10.04	11.80	9.96	11.86	9.89	11.81	9.83

Note: Standard errors in parentheses.

$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

When accounting for time-invariant team characteristics through team fixed effects, the relationship between payroll and wins remains significant but is slightly reduced ($\beta = 0.094$, $p < 0.001$). This suggests that some of the apparent effect of payroll in the baseline model is attributable to persistent team-specific factors such as organizational quality, market size, or management approach. The team fixed effects model substantially improves explanatory power ($R^2 = 0.406$), indicating that team-specific factors explain a considerable portion of the variance in win totals beyond what payroll alone can predict.

Figure 5 visualizes the team fixed effects from the linear model, illustrating the substantial heterogeneity in team performance after controlling for payroll differences. These effects represent each team's expected deviation in wins from the league average, holding payroll constant. Tampa Bay (TBR), Cleveland (CLE), and Houston (HOU) demonstrate the strongest positive fixed effects, suggesting these organizations consistently outperform expectations based on their payroll levels. Specifically, Tampa Bay shows approximately 11 wins above what would be predicted by payroll alone, while Cleveland and Houston achieve about 10 and 9 additional wins, respectively. Conversely, teams such as the Chicago White Sox (CHW), Detroit (DET), and Los Angeles Angels (LAA) exhibit the largest negative fixed effects, underperforming their payroll-predicted win totals by roughly 8-10 wins. Interestingly, the visualization reveals no clear pattern based on market size (indicated by color), with high-performing and low-performing organizations distributed across small, mid, and large markets. This reinforces the regression findings that team-specific factors such as likely reflecting organizational quality, player development systems, and front office efficiency play a crucial role in determining team success beyond mere spending capacity. The substantial magnitude of these fixed effects, ranging from approximately -10 to +11 wins, explains why the team fixed effects models demonstrated higher explanatory power than the baseline models that included only payroll.

The year fixed effects model, which controls for season-specific factors affecting all teams, yields a coefficient for total payroll ($\beta = 0.107$, $p < 0.001$) that is slightly higher than the baseline estimate. This suggests that accounting for year-to-year variations in the competitive environment slightly strengthens the estimated relationship between payroll and performance. However, the explanatory power of this model ($R^2 = 0.178$) is only marginally better than the baseline model, indicating that year effects alone add relatively little explanatory power.

The two-way fixed effects model, incorporating both team and year fixed effects, provides the most rigorous control for unobserved heterogeneity. In this specification, the estimated effect of payroll on wins remains strong and significant ($\beta = 0.106$, $p < 0.001$). This model has the highest explanatory power among the linear specifications ($R^2 = 0.414$), suggesting that accounting for both team-specific and year-specific factors provides the most comprehensive explanation of variation in team performance.

The baseline log-linear model examines how percentage changes in payroll relate to absolute changes in wins. The coefficient on the log transform of total payroll is strongly

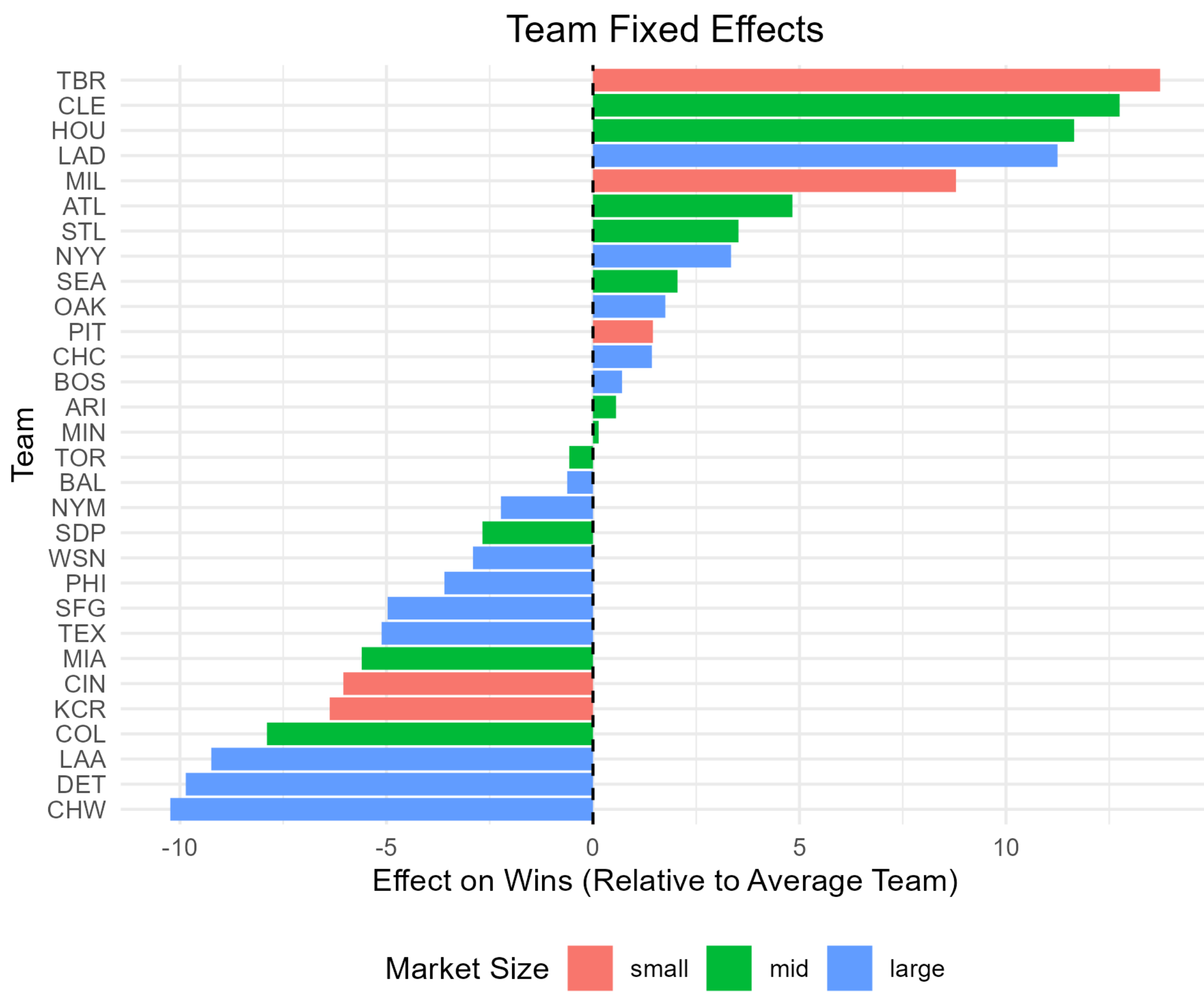

Figure 5

Team Fixed Effects from Eq 2

significant ($\beta = 12.764$, $p < 0.001$), indicating that a 1% increase in payroll is associated with approximately 0.128 additional wins. The model's intercept ($\beta_0 = 19.607$, $p < 0.05$) represents the expected wins for a team with a log-payroll of zero (which is not practically interpretable). The explanatory power of this model ($R^2 = 0.170$) is comparable to the linear baseline model.

When team fixed effects are incorporated into the log-linear specification, the coefficient on log-payroll increases to 13.120 ($p < 0.001$). This suggests that after accounting for persistent team differences, the elasticity of wins with respect to payroll is somewhat higher than in the pooled model. The explanatory power of this model ($R^2 = 0.423$) substantially exceeds the baseline log-linear model and is slightly higher than its linear counterpart with team fixed effects.

The year fixed effects log-linear model yields a coefficient on log-payroll ($\beta = 13.240$, $p < 0.001$) that is slightly higher than the baseline log-linear estimate. Similar to the linear specifications, accounting for year effects alone provides only modest improvement in explanatory power ($R^2 = 0.177$) compared to the baseline model.

The two-way fixed effects log-linear model combines both team and year fixed effects and shows the strongest relationship between log-payroll and wins ($\beta = 14.223$, $p < 0.001$). This indicates that after controlling for both team-specific and year-specific factors, a 1% increase in payroll is associated with approximately 0.142 additional wins. This model has the highest explanatory power among all specifications ($R^2 = 0.430$), suggesting that the log-linear two-way fixed effects model best captures the relationship between team spending and performance.

The model fit statistics provide several insights into the relationship between payroll and team performance. First, models with team fixed effects consistently outperform those without team fixed effects in terms of explanatory power (R^2) and fit metrics (AIC, RMSE). This highlights the importance of persistent team-specific factors in determining performance beyond simple payroll levels.

Second, the log-linear specifications generally show slightly better fit than their linear counterparts across most metrics. The log-linear team fixed effects model has the lowest AIC (2065.6) among all specifications, while the log-linear two-way fixed effects model has the lowest RMSE (9.83). This suggests that the relationship between payroll and performance may be better characterized as proportional rather than strictly linear.

Third, the substantial increase in R^2 when moving from baseline models ($R^2 \approx 0.17$) to fixed effects models ($R^2 \approx 0.41 - 0.43$) indicates that while payroll is an important determinant of team success, team-specific factors explain a larger portion of the variance in performance. The adjusted R^2 values, which penalize additional parameters, remain substantially higher for the fixed effects models despite the large number of additional parameters.

Overall, these results confirm a strong, positive relationship between team payroll and regular season wins, with evidence that this relationship is robust to various model specifications and controls for unobserved heterogeneity. The consistency of the payroll coefficient across specifications suggests that the relationship is not just due to team-specific or time-specific confounding factors.

Payroll Allocation and Market Size Effects

Table 2 explores how different components of payroll, pre-arbitration, arbitration, and veteran salary shares, relate to both total wins and win variability corresponding to Equations 9–16.

The baseline win model examined the direct relationship between payroll allocation percentages and team wins without additional controls. The regression is estimated without an intercept, so the coefficients represent the expected number of wins a team would achieve if 100% of payroll were allocated to a single category. The largest coefficient belongs to arbitration-eligible payroll percentage ($\beta = 97.410$, $p < 0.001$), indicating that a team allocating all its payroll to arbitration-eligible players is predicted to win approximately 97 games. This is followed by veteran payroll percentage ($\beta = 83.488$, $p < 0.001$), suggesting a similarly strong but slightly smaller payoff. The coefficient for pre-arbitration payroll percentage is smaller in magnitude ($\beta = 22.133$, $p < 0.05$), implying that teams relying solely on pre-arbitration players would expect significantly fewer wins. The substantially larger coefficient for arbitration payroll percentage suggests that, holding other allocations constant, investment in arbitration-eligible players yields the greatest marginal returns for team wins.

When controlling for total team payroll, the estimated win values associated with different payroll allocations shifted, but remained significant. If a team with fixed total

Table 2

Effect of Payroll Allocations on Wins and Win Variability

	Wins				Win Variability			
	Baseline	Total Payroll	Market Size	Payroll Interaction	Baseline	Total Payroll	Market Size	Payroll Interaction
pre_arb_payroll_pct	22.133*	46.714***	21.822*	50.559*	14.720***	12.315***	16.769***	9.556
	(8.896)	(9.621)	(9.291)	(19.739)	(3.152)	(3.576)	(3.257)	(7.335)
arb_payroll_pct	97.410***	86.538***	96.382***	90.276***	6.688***	7.752***	8.286***	6.884
	(4.619)	(4.838)	(5.016)	(12.236)	(1.636)	(1.798)	(1.759)	(4.547)
vet_payroll_pct	83.488***	62.881***	83.250***	61.984***	8.162***	10.178***	8.292***	10.294***
	(1.825)	(4.208)	(1.864)	(5.618)	(0.647)	(1.564)	(0.654)	(2.088)
total_payroll_m		0.102***		0.015		-0.010		0.045
		(0.019)		(0.301)		(0.007)		(0.112)
market_sizesmall			0.242					-1.773*
			(2.265)					(0.794)
market_sizemid			1.111					-1.096+
			(1.686)					(0.591)
arb_payroll_pct × total_payroll_m				0.059				-0.049
				(0.348)				(0.129)
vet_payroll_pct × total_payroll_m				0.099				-0.058
				(0.301)				(0.112)
Num.Obs.	270	270	270	270	270	270	270	270
R2	0.978	0.981	0.978	0.981	0.796	0.798	0.801	0.798
R2 Adj.	0.978	0.980	0.978	0.980	0.794	0.795	0.797	0.794
AIC	2118.7	2092.8	2122.2	2096.5	1558.4	1558.4	1556.2	1562.0
RMSE	12.06	11.45	12.05	11.45	4.27	4.26	4.22	4.25

Note: + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

payroll allocated all spending to pre-arbitration players, the expected win total would be approximately 47 games ($\beta = 46.714$, $p < 0.001$) which is more than double the baseline estimate. This indicates that within-budget investments in younger, lower-cost players can yield competitive performance when total spending is held constant. The predicted win total for full arbitration-eligible allocation remained high at 87 wins ($\beta = 86.538$, $p < 0.001$), only modestly lower than in the baseline model. In contrast, the win return for veteran payroll allocation dropped more noticeably to 63 wins ($\beta = 62.881$, $p < 0.001$), suggesting that part of the veteran effect observed in the baseline may reflect their tendency to be employed by high-spending teams, rather than purely their individual value. These results reinforce the efficiency of arbitration-level spending relative to other payroll tiers, particularly under constrained budgets. Total payroll itself showed a significant positive relationship with wins ($\beta = 0.102$,$p < 0.001$), confirming the expected relationship between financial investment and on-field success. Specifically, each additional million dollars in total payroll was associated with approximately 0.1 additional wins, holding allocation percentages constant. This estimate is comparable to the estimates from in the regressions from Table 1.

In the market size control model, the coefficients for pre-arbitration payroll percentage ($\beta = 21.822$, $p < 0.05$), arbitration payroll percentage ($\beta = 96.382$, $p < 0.001$), and veteran payroll percentage ($\beta = 83.250$, $p < 0.001$) remained largely consistent with the baseline model. Neither small market ($\beta = 0.242$, $p > 0.05$) nor mid-market ($\beta = 1.111$, $p > 0.05$) indicators showed significant relationships with wins compared to large markets, suggesting team success is more closely tied to how resources are allocated rather than market size advantage.

The total payroll interaction model examined whether the effectiveness of different allocation strategies varies with overall spending levels. The main effects for pre-arbitration ($\beta = 50.559$, $p < 0.05$), arbitration ($\beta = 90.276$, $p < 0.001$), and veteran ($\beta = 61.984$, $p < 0.001$) payroll percentages remained significant. However, the interaction terms between payroll allocation percentages and total payroll were not statistically significant (arbitration $\times$ total payroll: $\beta = 0.059$, $p > 0.05$; veteran $\times$ total payroll: $\beta = 0.099$, $p > 0.05$), indicating that the effectiveness of allocation strategies does not meaningfully vary with total payroll level. This suggests that optimal allocation strategies may be relatively consistent across different spending levels.

To assess the stability of team performance, the relationship between payroll allocation strategies and win variability was analyzed. This provides insight into which allocation strategies lead to more consistent performance outcomes.

In the baseline model of win variability, the estimated coefficients reflect the expected level of performance volatility if a team allocated its entire payroll to a single category. The highest coefficient belongs to pre-arbitration payroll ($\beta = 14.720$, $p < 0.001$), indicating that teams composed entirely of pre-arbitration players would exhibit the greatest variability in win outcomes. While both arbitration payroll ($\beta = 6.688$, $p < 0.001$) and veteran payroll ($\beta = 8.162$, $p < 0.001$) also show significant positive associations with variability, their smaller magnitudes suggest comparatively more stable performance profiles. These results imply that heavy reliance on inexperienced, low-cost players introduces greater uncertainty in team outcomes, whereas more experienced (and expensive) rosters offer relatively more predictable performance.

When controlling for total payroll, the relationship between pre-arbitration allocation and win variability remained strong ($\beta = 12.315$, $p < 0.001$), though slightly reduced compared to the baseline model. Arbitration payroll percentage ($\beta = 7.752$, $p < 0.001$) and veteran payroll percentage ($\beta = 10.178$, $p < 0.001$) maintained significant positive associations with variability. Interestingly, the veteran coefficient increased in this specification, suggesting that controlling for total payroll reveals a stronger relationship between veteran-heavy rosters and performance inconsistency. Total payroll itself did not significantly impact win variability ($\beta = -0.010$, $p > 0.05$), indicating that higher spending alone does not lead to more consistent performance outcomes.

In the market size control model, all three allocation percentages maintained their significant positive relationships with win variability (pre-arbitration: $\beta = 16.769$, $p < 0.001$; arbitration: $\beta = 8.286$, $p < 0.001$; veteran: $\beta = 8.292$, $p < 0.001$). Notably, small market size showed a significant negative association with win variability ($\beta = -1.773$, $p < 0.05$), suggesting smaller market teams experience less performance volatility than large market teams. Mid-sized markets also showed a marginally significant negative relationship with win variability ($\beta = -1.096$, $p < 0.1$). These findings suggest that market constraints may lead teams to adopt more consistent, if potentially less optimal, strategies.

In the total payroll interaction model, pre-arbitration allocation's relationship with win variability became non-significant ($\beta = 9.556$, $p > 0.05$), while veteran allocation maintained the strongest relationship ($\beta = 10.294$, $p < 0.001$). The interaction effects between allocation percentages and total payroll were not statistically significant for win variability, though they trended negative (arbitration × total payroll: $\beta = -0.049$, $p > 0.05$; veteran × total payroll: $\beta = -0.058$, $p > 0.05$). This suggests that the relationship between allocation strategies and performance consistency is relatively uniform across different spending levels.

All four win models demonstrated exceptionally strong explanatory power, with R^2 values ranging from 0.978 to 0.981, indicating that these specifications explain approximately 98% of the variance in team wins. The adjusted R^2 values (0.978 to 0.980) remained nearly identical to the unadjusted values, suggesting that the additional parameters in the more complex models are contributing meaningfully to the explanatory power.

The win variability models showed good but comparatively lower explanatory power than the win models, with R^2 values between 0.796 and 0.801, explaining approximately 80% of the variance in performance consistency. The market size model demonstrated the best fit according to RMSE (4.22), while the total payroll model had the lowest AIC (1558.4, tied with the baseline model). The differences in AIC values across the win variability models were minimal, suggesting that the additional predictors in the more complex models contribute less to explaining variance in win variability than they do for explaining variance in wins.

Overall, the model fit metrics suggest that payroll allocation percentages are strongly predictive of both team wins and win variability, with total payroll adding significant explanatory power for wins but less so for win variability. Market size appears to be more relevant for explaining differences in performance consistency than for explaining differences in overall performance level.

Conclusion

This analysis demonstrates that while market size and payroll influence team success in Major League Baseball, they do not solely determine outcomes. Smaller market teams can achieve success comparable to that of larger market teams, though they often do so through different roster construction strategies. Higher total payroll is generally associated with better regular season performance, but the returns diminish as spending increases, suggesting inefficiencies in excessive payroll investments.

Importantly, team-specific factors such as management decisions, player development systems, and strategic resource allocation account for a substantial portion of win variation beyond payroll alone. This indicates that strong organizational strategies can help financially constrained teams remain competitive.

From a value perspective, arbitration-eligible players provide the most efficient return on investment in terms of wins. These players tend to be in the prime years of their career and are less expensive than players obtained through free-agency. Pre-arbitration players contribute less consistently and are associated with higher variability in team success, while veteran players add wins but with increased volatility and cost. Pre-arbitration players are young and inexperienced which can explain their smaller contributes to win totals and increased contribution to win variability. Veteran players have more experience which can help explain there contribution to wins, but these players come with higher risk than arbitration players as they are older which can increase their risk of injury or decline in performance.

Overall, while payroll remains a meaningful factor in team performance, strategic spending and effective organizational management allow teams to overcome financial disparities and remain competitive.

While this study offers a comprehensive view of how payroll structure affects regular season outcomes, several avenues remain for further research. One important extension would be to examine the impact of roster construction and payroll allocation on postseason performance. Regular season success does not always translate to playoff wins, and understanding how different team-building approaches fare in high-stakes environments could yield valuable insights.

Future work could also benefit from incorporating player-level data, including individual performance metrics, positions, and durability. This would allow for more nuanced assessments of value by position or role and could help identify more specific types of players who offer the most cost-effective contributions.

Moreover, this analysis includes only players who appeared in at least one game during the season, excluding those on the payroll who were injured or remained in the minors for the entire season. Accounting for these hidden costs in future models could produce a more accurate estimate of how financial resources are actually utilized. Additionally, the ever changing nature of a team's Major League roster makes it hard to capture the team's roster in a season. Further analysis could examine the construction of the roster during certain points in the season to get a greater understanding of a team's roster building strategies.

Ultimately, this study underscores that while financial investment is a key component of competitive success in Major League Baseball, it is not the sole determinant. Strategic allocation of payroll, especially through maximizing the value of arbitration-eligible players and balancing the risks associated with veterans and pre-arbitration talent, can offer a path to success for teams across market sizes. As the game continues to evolve through changes in player development, analytics, and labor dynamics, understanding the nuanced relationship between money and performance will remain essential for front offices striving to build winning teams in an increasingly competitive environment.

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Appendix A

MLB Team Market Size Classifications

Team	Team ID	League	Division	Market Size	Market Tier
Arizona Diamondbacks	ARI	NL	West	3,251,876	Mid
Oakland Athletics	OAK	AL	West	7,039,362	Large
Atlanta Braves	ATL	NL	East	4,112,198	Mid
Baltimore Orioles	BAL	AL	East	7,608,070	Large
Boston Red Sox	BOS	AL	East	5,819,100	Large
Chicago Cubs	CHC	NL	Central	9,157,540	Large
Chicago White Sox	CHW	AL	Central	9,157,540	Large
Cincinnati Reds	CIN	NL	Central	1,979,202	Small
Cleveland Guardians	CLE	AL	Central	2,945,831	Mid
Colorado Rockies	COL	NL	West	2,581,506	Mid
Detroit Tigers	DET	AL	Central	5,456,428	Large
Houston Astros	HOU	AL	West	4,669,571	Mid
Kansas City Royals	KCR	AL	Central	1,776,062	Small
Los Angeles Angels	LAA	AL	West	16,373,645	Large
Los Angeles Dodgers	LAD	NL	West	16,373,645	Large
Miami Marlins	MIA	NL	East	3,878,380	Mid
Milwaukee Brewers	MIL	NL	Central	1,689,572	Small
Minnesota Twins	MIN	AL	Central	2,968,806	Mid
New York Mets	NYM	NL	East	21,199,865	Large
New York Yankees	NYY	AL	East	21,199,865	Large
Philadelphia Phillies	PHI	NL	East	6,188,463	Large
Pittsburgh Pirates	PIT	NL	Central	2,358,695	Small
San Diego Padres	SDP	NL	West	2,813,833	Mid
San Francisco Giants	SFG	NL	West	7,039,362	Large
Seattle Mariners	SEA	AL	West	3,554,760	Mid
St. Louis Cardinals	STL	NL	Central	2,603,607	Mid
Tampa Bay Rays	TBR	AL	East	2,395,997	Small
Texas Rangers	TEX	AL	West	5,221,801	Large
Toronto Blue Jays	TOR	AL	East	4,682,897	Mid
Washington Nationals	WSN	NL	East	7,608,070	Large

Table A1

MLB Teams by Market Size Designation ("Baseball Markets," n.d.)

Appendix B

Code Availability

The complete source code for this project is available at the following GitHub repository:

`https://github.com/selah-dean/econ-399`